Changelog: PerfAudit v1.9

Release Date: November 9, 2026 | Build Type: Production / Audit-Ready

[NEW] Passive Game FPS Auditing

- **Passive ETW Frame Capture:** Added game FPS logging utilizing kernel Event Tracing for Windows (ETW) subscribed to authoritative DXGI Present start events (IDs 42 and 55). Operates purely out-of-process with zero DLL injection, API hooking, or overlays, ensuring passive, non-intrusive monitoring.
- **Dual Target Architecture:** Added a second dropdown "Select Game (FPS)" with a "None" option. CPU sampling and game FPS logging can operate completely independently or concurrently in parallel on a shared, synchronized high-resolution timebase.
- **Independent Logger Resiliency:** If one target process terminates during an active session (e.g., game exit), the remaining logger continues monitoring uninterrupted.
- **Real-Time FPS Telemetry:** Added a live "FPS: 0" status label alongside the CPU usage display, updating at 1 Hz with zero overhead on sampling cadence.

[NEW] Single Multi-Worksheet Excel (.xlsx) Report

- **OpenXML Workbook Generation:** Replaced CSV export with a unified, professional Excel (.xlsx) workbook per run generated via EPPlus 4.5.3.3.
- **Sheet "CPU_Usage":** Preserves the forensic 100ms sample log, millisecond timestamps (' HH:mm:ss.fff), burst threshold indicators, and static summary header/footer rows of prior audit baselines.
- **Sheet "CPU_Graph":** Dedicated analytical sheet featuring whole-run summary statistics (Average CPU %, Peak CPU %, Threshold used %, Total burst samples, Seconds containing a burst, Duration [mm]:ss), a per-second peak CPU vs. time table, and an embedded DrawingML line chart plotting peak CPU % per second, a dashed threshold line, and discrete red circle markers for burst seconds (empty seconds plotted as gaps).
- **Sheet "FPS_Log":** Per-second audit log with one row per full second recording Timestamp (HH:mm:ss, start of that second), target game Process, time_s (fraction of a day formatted [mm]:ss), avg_fps, and low_1pct_fps. Individual frametimes are maintained in memory only as needed for calculations.
- **Sheet "FPS_Graph":** Streamlined performance analytics sheet providing session summary metrics (Average FPS, Average frametime (ms), and 1% low FPS (percentile)) and a single embedded DrawingML line chart plotting Average FPS and 1% low over time directly referencing FPS_Log.

[UI/UX] Incremental Type-Ahead Search

- **Shared Prefix Search Helper:** Integrated an incremental type-ahead search helper into both "Select Process (CPU)" and "Select Game (FPS)" ComboBoxes without altering native dropdown styling.
- **Progressive Matching:** Typing characters incrementally narrows and selects the first matching item by displayed process name without resetting between keystrokes.
- **Keyboard Controls:** Backspace removes the last character and re-evaluates the match; Escape clears the search buffer. An automatic 1.5-second inactivity timeout resets the buffer, and buffers clear on list refresh or focus loss while leaving native mouse and arrow navigation untouched.

[DEPENDENCIES & PRIVILEGES]

- **Administrator Privilege Requirement:** Creation and management of ETW trace sessions requires elevated privileges. Configured application manifest to requireAdministrator.
- **Managed NuGet Dependencies:** Integrated EPPlus 4.5.3.3 (LGPL 2.1) for zero-dependency spreadsheet generation and Microsoft.Diagnostics.Tracing.TraceEvent 3.1.8 for kernel ETW event consumption.